Correlation

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Correlation

- In real life situations, especially in social sciences and in business, we often to know whether two or more variables are related, and if so, how they are related.

- The following are some questions of interest,

1. Is there a relationship between two or more variables?
2. If so, what is the relationship?

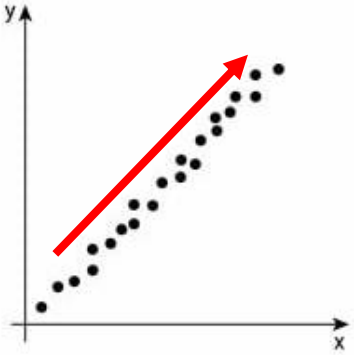
Correlation

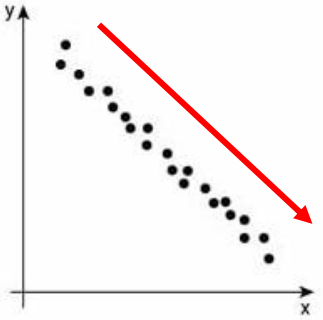
- Relationship between two or more variables
- Which gives us strength or degree and direction of association

Correlation:

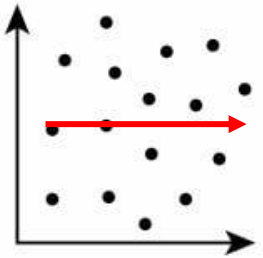
Relationship between two or more variables which gives us strength or degree and direction of association

Types of Correlation





No Correlation



On the basis of direction

- Positive correlation
- Negative correlation

On the basis of variables

On the basis of linearity

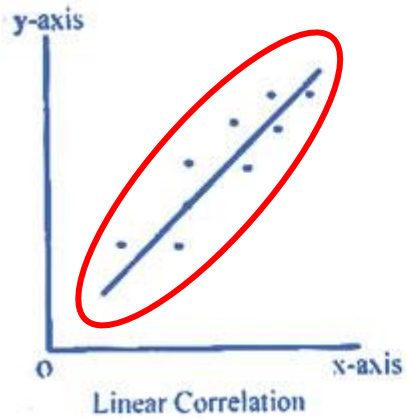
Types of Correlation

On the basis of direction	• Positive correlation • Negative correlation
On the basis of variables	• Simple correlation • Multiple correlation
On the basis of linearity	

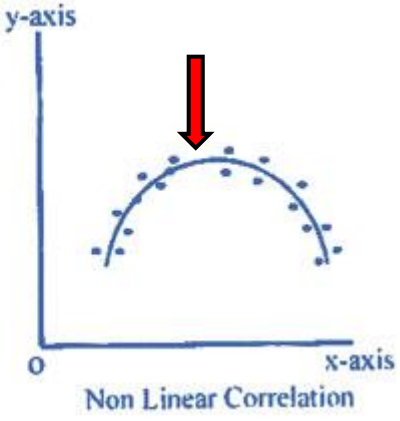
Types of Correlation

On the basis of direction	• Positive correlation • Negative correlation
On the basis of variables	• Simple correlation (Two variables) • Multiple correlation (More than two)
On the basis of linearity	

Types of Correlation



Linear Correlation



Non Linear Correlation

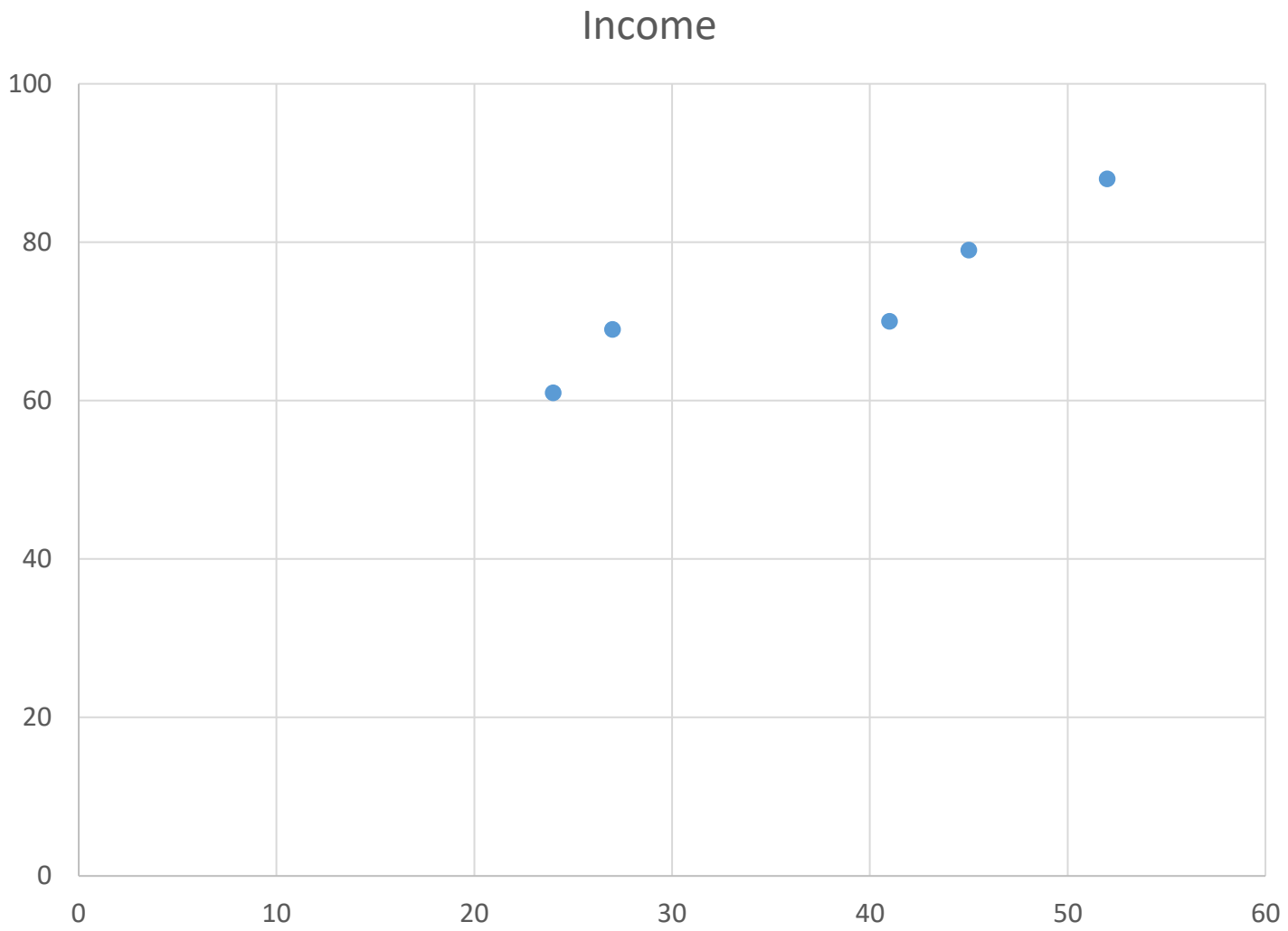
On the basis of direction	• Positive correlation • Negative correlation
On the basis of variables	• Simple correlation • Multiple correlation
On the basis of linearity	• Linear correlation • Non-Linear correlation

Measures

- Correlation can be measured by

1. Scatter Diagram
2. Karl Pearson's Correlation Coefficient
3. Spearman Rank Correlation Coefficient
4. Method of Least Square

Scatter Diagram



With the increase of age, the Glucose Level also increased. Thus, there is a positive correlation between “Age” and “Glucose Level”.

Limitations

- Accurate degree and strength of correlation can not be obtained by scatter diagram.

Karl Pearson's Correlation

- Both variables are measured in interval or ratio scales
- Relationship between variables is linear
- Denoted by ***r***

Karl Pearson's Correlation

$$r = \frac{\sum(x_i - \bar{x})\,(y_i - \bar{y})}{\sqrt{\sum(x_i - \bar{x})^2}\,\sqrt{\sum(y_i - \bar{y})^2}}$$

$$\boldsymbol{r = \frac{\sum x_i y_i - \frac{\sum x_i \sum y_i}{n}}{\sqrt{\sum x_i^2 - \frac{(\sum x_i)^2}{n}}\,\sqrt{\sum y_i^2 - \frac{(\sum y_i)^2}{n}}}}$$

Range of r is [-1 to +1]

If $r = +1$; Perfect positive correlation

If $r = -1$; Perfect negative correlation

If $r = 0$; No correlation

Correlation	Negative	Positive
Weak	-0.29 to -0.10	0.10 to 0.29
Medium	-0.49 to -0.30	0.30 to 0.49
Moderate	-0.50 to -0.79	0.50 to 0.79
Strong	-1.00 to -0.80	0.80 to 1.00

Karl Pearson's Correlation

- Example: Following are the Advertising Expenditure and Sales (in 000$) Revenue of 10 months of a company.

Expenditure	50	62	57	58	65	60	56	63	57	59
Sales	420	650	630	530	540	600	610	550	540	590

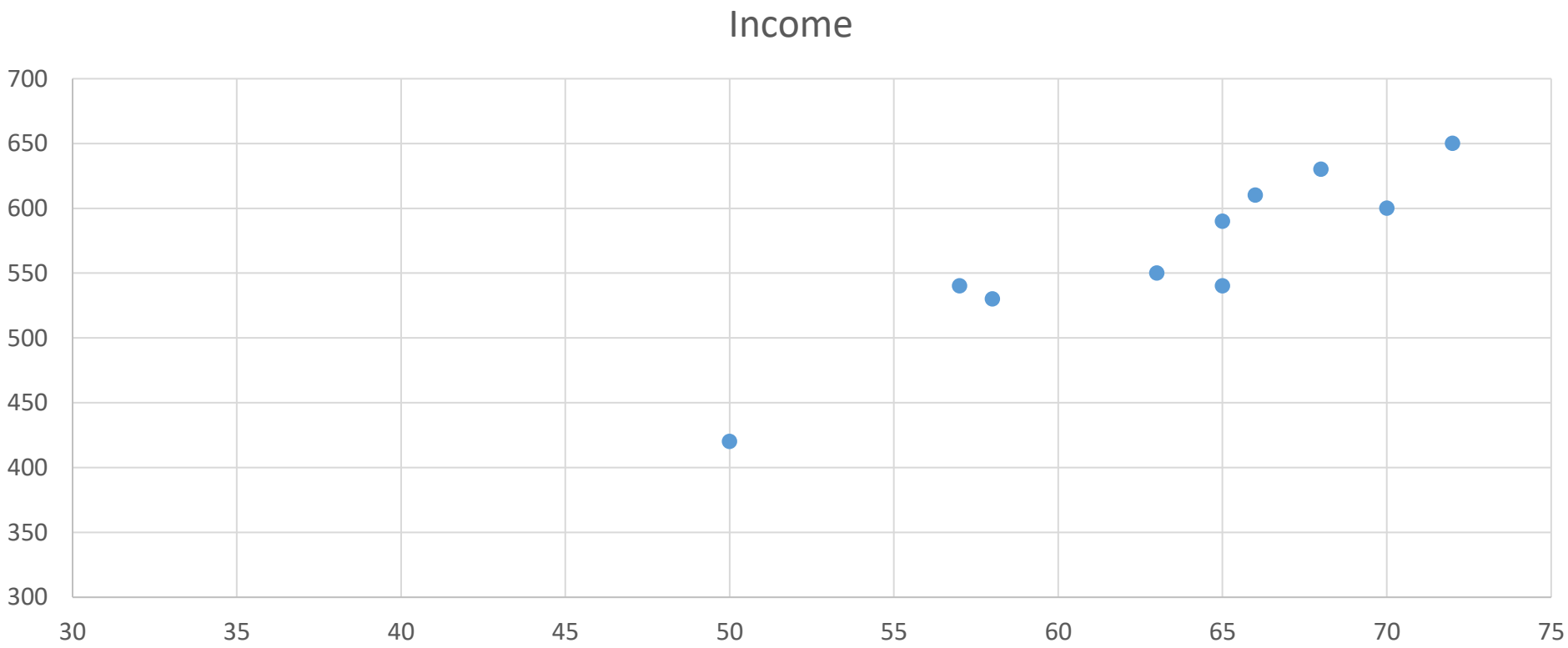
a) Draw the scatter plot

b) Compute the correlation coefficient value with proper interpretation

Karl Pearson's Correlation

Example: Following are the Advertising Expenditure and Sales (in 000$) Revenue of 10 months of a company :

a) Draw the scatter plot



Cont.

$$r = \frac{\sum x_i y_i - \frac{\sum x_i \sum y_i}{n}}{\sqrt{\sum x_i^2 - \frac{(\sum x_i)^2}{n}} \sqrt{\sum y_i^2 - \frac{(\sum y_i)^2}{n}}}$$

x(Expenditure)	y (Sales)	x_i^2	y_i^2	$x_i \times y_i$
50	420	2500	176400	21000
62	650	3844	422500	40300
57	630	3249	396900	35910
58	530	3364	280900	30740
65	540	4225	291600	35100
60	600	3600	360000	36000
56	610	3136	372100	34160
63	550	3969	302500	34650
57	540	3249	291600	30780
59	590	3481	348100	34810
$\sum x_i = 587$	$\sum y_i = 5660$	$\sum x_i^2 = 34617$	$\sum y_i^2 = 3242600$	$\sum x_i y_i = 333450$

Cont.

$$r = \frac{\sum x_i y_i - \frac{\sum x_i \sum y_i}{n}}{\sqrt{\sum x_i^2 - \frac{(\sum x_i)^2}{n}} \sqrt{\sum y_i^2 - \frac{(\sum y_i)^2}{n}}} = \frac{333450 - \frac{587 \times 5660}{10}}{\sqrt{34617 - \frac{(587)^2}{10}} \sqrt{3242600 - \frac{(5660)^2}{10}}} = ?$$

Self Practice

- Following are the height and weight of 10 students.

Height	62	72	68	58	65	70	66	63	60	72
Weight	50	65	63	50	54	60	61	55	54	65

Properties

1. Correlation coefficient has no unit.
2. The sign of correlation coefficient gives the direction of the association.
3. The correlation coefficient is between -1 and $+1$.
4. Correlation coefficient is a symmetric measure, $i.e., r_{xy} = r_{yx}$
5. Correlation is sensitive to extreme observations.

Thank You